Newton School of Technology, Rishihood University

B.Tech Computer Science and Artificial Intelligence

Project Report

Card-Fraud Triage: GMM Density Scoring vs Autoencoder Anomaly Detector

Phase 1: Machine Learning Evaluation

Gaussian Mixture density scoring of legitimate card transactions

Project 26 | Team ID: T068 | Section: B

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Abstract

Card fraud is rare (0.167% of transactions in our data), its patterns change over time, and confirmed labels arrive only weeks after a transaction. A practical fraud system is therefore a *triage* tool. It ranks transactions so that a small team of analysts reviews the most suspicious ones first. In this first phase we frame fraud detection as *novelty detection*. A Gaussian Mixture Model (GMM), fitted by the Expectation–Maximisation (EM) algorithm, learns the probability density of *legitimate* transactions only, without ever seeing a fraud label, and the negative log-likelihood $-\log p(\mathbf{x})$ serves as the anomaly score.

On the ULB credit-card dataset (284,807 transactions) we use a strictly chronological, leakage-controlled protocol: duplicates are removed before splitting, the train/validation/test blocks are ordered in time, models are fitted on legitimate rows only, and thresholds are chosen on validation and frozen for the test period. We find three things:

1. **Feature engineering matters most.** A per-feature quantile (Gaussianising) transform is the decisive design choice. It turns a single-Gaussian detector from PR-AUC 0.163 into 0.691 and removes most of EM's seed-to-seed instability.
2. **More components do not mean better detection.** AIC and BIC keep asking for more mixture components ($K = 24$), yet $K = 1$ to 4 detect fraud as well or better, and large mixtures are unstable.
3. **The final model is a strong triage tool.** The selected GMM ($K = 4$) reaches a test PR-AUC of 0.686 (95% CI 0.572–0.808). At a false-positive budget of 0.1% of legitimate customers it catches 75.7% of test frauds, and 90% of its top-50 alerts are fraud. This comes close to a supervised model that uses labels (0.739).

We document the false-positive trade-offs, an illustrative cost analysis and the limitations that motivate the deep learning phase.

Keywords: fraud detection, anomaly detection, novelty detection, Gaussian mixture model, expectation–maximisation, model selection, imbalanced evaluation, precision–recall.

Contents

Abstract **1**

1 Introduction **4**

1.1 Background and motivation 4

1.2 Problem statement 4

1.3 Objectives 4

1.4 Contributions 5

1.5 Organisation of the report 5

2 Literature Review **5**

2.1 Fraud detection as a learning problem 5

2.2 Anomaly and novelty detection 5

2.3 Gaussian mixtures, EM and model order 6

2.4 Evaluation under extreme class imbalance 6

2.5 Summary and research gap 6

3 Dataset and Exploratory Data Analysis **7**

3.1 Dataset description 7

3.2 Data quality 7

3.3 Class imbalance 7

3.4 Time of day and transaction amount 8

3.5 Which components separate the classes? 8

3.6 Is legitimate behaviour Gaussian? 9

3.7 Correlation structure 10

3.8 Chronological split and base rates 10

4 Feature Engineering **11**

4.1 Engineered features 11

4.2 Scaling options 12

5 Methodology **12**

5.1 Overview . . . 12
5.2 Gaussian Mixture Model . . . 13
5.3 Expectation–Maximisation . . . 13
5.4 Anomaly score and decision rule . . . 14
5.5 Model selection with AIC and BIC . . . 14
5.6 Computational complexity . . . 14
5.7 Reference baselines . . . 15
5.8 Evaluation protocol and metrics . . . 15
5.9 Experimental setup . . . 16

6 Experiments and Results **16**
6.1 Feature-engineering ablation . . . 16
6.2 Choosing K: information criteria versus detection quality . . . 17
6.3 Main results . . . 18
6.4 Score distribution . . . 19
6.5 Threshold analysis and false-positive trade-offs . . . 19
6.6 Cost-based operating point . . . 20
6.7 Stability: the high-dimensional gap, measured . . . 21
6.8 Error analysis . . . 22

7 Discussion **22**
7.1 Interpretation . . . 22
7.2 The gap that motivates Phase 2 . . . 22
7.3 Limitations and threats to validity . . . 23

8 Conclusion and Future Work **23**

A Reproducibility **25**

B Team Contributions **25**

1 Introduction

1.1 Background and motivation

Every card payment must be approved within milliseconds, yet only a few hundred out of hundreds of thousands are fraudulent. Banks cannot investigate every transaction. Instead, a fraud system raises *alerts*, and a limited team of analysts reviews them, blocks cards, or contacts customers [1]. The key question is therefore not "is this transaction fraud?" but "*which few transactions should analysts look at first?*". This is a ranking (triage) problem.

Supervised classifiers need many labelled frauds. In practice labels are scarce, arrive late (after chargebacks), and describe only *past* fraud patterns [2]. Legitimate transactions, by contrast, are abundant and available immediately. This motivates *novelty detection*: learn a model of normal behaviour and flag departures from it [3, 4].

1.2 Problem statement

The project brief asks us to frame card fraud as anomaly detection. Models are fitted primarily on legitimate transactions, and probability-density scores are compared with reconstruction error on a held-out mixed set. In Phase 1 (machine learning) we must:

- fit a Gaussian Mixture Model on legitimate transactions, tune the component count with AIC and BIC, and use low log-likelihood as the anomaly score;
- use a chronological split, fit scaling and the GMM only on legitimate training rows, keep the fraud label hidden during training, select thresholds on validation data, and report base rates;
- report PR-AUC, recall at fixed false-positive budgets and precision at a review capacity, and *not* rely on accuracy;
- identify the gap of the GMM that a deep-learning model (Phase 2) should address.

1.3 Objectives

1. Build a reproducible, leakage-free evaluation protocol shared by every model of both phases.
2. Understand the data: quality, imbalance, temporal structure, separability and how Gaussian it is.
3. Engineer features and preprocessing that suit a Gaussian density model.
4. Fit, select and validate the GMM, and compare it with meaningful reference baselines.
5. Translate scores into operating policies (thresholds, review workload, cost) for fraud analysts.
6. Measure the GMM's weaknesses quantitatively, to motivate Phase 2.

1.4 Contributions

- A leakage-controlled protocol: deduplication, a chronological 60/20/20 split, legitimate-only fitting, and thresholds frozen on validation.
- A feature-engineering study showing that *marginal Gaussianisation* (quantile transform) is what makes a GMM work on heavy-tailed transaction features.
- An analysis of AIC/BIC model selection at large sample size, where the criteria and detection quality disagree.
- A quantified stability study of EM restarts, measuring the "instability in high dimensions" gap named in the brief.
- A triage-oriented evaluation with operating points, workload per hour and an illustrative cost analysis.

1.5 Organisation of the report

Section 2 reviews related work. Sections 3 and 4 analyse the data and describe the feature engineering. Section 5 presents the GMM, the EM algorithm, model selection and the evaluation protocol. Section 6 reports the experiments. Section 7 discusses the findings and limitations, and Section 8 concludes. The appendices cover reproducibility and team contributions.

2 Literature Review

2.1 Fraud detection as a learning problem

Bolton and Hand [2] review statistical fraud detection. They distinguish *supervised* methods, which learn from labelled past fraud, from *unsupervised* methods, which flag departures from normal behaviour. They argue that unsupervised methods are essential because fraudsters adapt and labels are delayed and incomplete. Dal Pozzolo *et al.* released the dataset we use [5] and later described a realistic fraud-detection system in which investigators review a fixed number of alerts per day [1]. In that setting the relevant metric is precision among the top-k alerts, which is why we report precision at a review capacity. Carcillo *et al.* [6] showed that unsupervised outlier scores add information to supervised fraud classifiers, which supports treating anomaly scores as triage signals.

2.2 Anomaly and novelty detection

Chandola *et al.* [3] classify anomaly-detection techniques as statistical (density-based), distance-based, clustering-based, classification-based and spectral. Statistical techniques assume that normal instances occur in high-probability regions of a stochastic model, which is exactly the GMM score $-\log p(\mathbf{x})$. Pimentel *et al.* [4] review *novelty detection* (training on normal data only). They identify GMM-based density estimation as the canonical probabilistic approach and note its sensitivity to dimensionality and to the choice of model order. Isolation Forest [7]

isolates anomalies with random axis-aligned splits and needs no density model; we use it as a non-density reference.

2.3 Gaussian mixtures, EM and model order

Dempster, Laird and Rubin [8] formalised the EM algorithm for maximum-likelihood estimation with latent variables and proved that each iteration does not decrease the likelihood. McLachlan and Peel [9] and Bishop [10] give the standard treatment of finite mixtures: covariance structures (full, diagonal, tied), the singularity problem when a component collapses onto a few points, and covariance regularisation. To choose the number of components, Akaike's information criterion (AIC) [11] and Schwarz's Bayesian information criterion (BIC) [12] penalise the log-likelihood by model size. BIC is consistent for the true order *when the model family is correct*, an assumption that heavy-tailed real data violates.

2.4 Evaluation under extreme class imbalance

Davis and Goadrich [13] show that a ROC curve can look excellent while precision is poor. Saito and Rehmsmeier [14] show empirically that precision–recall (PR) plots are more informative than ROC plots on imbalanced data. Our primary metric is therefore PR-AUC (average precision). We never report accuracy: a model that flags nothing is 99.83% "accurate".

2.5 Summary and research gap

Table 1: Summary of the literature that shapes our Phase-1 design.

Work	Contribution	How we use it
Bolton & Hand [2]	Unsupervised fraud detection is needed: labels are scarce and late	Legitimate-only training protocol
Dal Pozzolo *et al.* [1, 5]	Dataset; analysts review a fixed number of alerts	Precision at review capacity (P@k)
Dempster *et al.* [8]	EM algorithm for latent-variable models	GMM fitting; local optima studied
Akaike [11], Schwarz [12]	AIC / BIC model selection	Sweep of K and covariance type
Pimentel *et al.* [4]	GMM novelty detection is sensitive to dimension and order	The "gap" we measure
Liu *et al.* [7]	Isolation Forest	Non-density reference baseline
Saito & Rehmsmeier [14]	PR beats ROC under imbalance	PR-AUC as the primary metric

Most published work on this dataset trains *supervised* classifiers with random (not time-ordered) splits and reports accuracy or ROC-AUC. Fewer studies evaluate a label-free density model under a chronological, leakage-free protocol with triage metrics, or examine how preprocessing and model order affect the stability of EM. That is the focus of this phase.

3 Dataset and Exploratory Data Analysis

3.1 Dataset description

We use the Credit Card Fraud Detection dataset released by the ULB Machine Learning Group [5] (distributed via Kaggle). It contains 284,807 transactions made by European cardholders over two days in September 2013, of which 492 are fraudulent. Table 2 describes the columns. The notebooks download the file from a public mirror and verify its SHA-256 checksum, so every run uses identical data.

Table 2: Dataset columns.

Column	Type	Meaning
`Time`	numeric	Seconds elapsed since the first transaction in the file (0 to 172,792 s ≈ 48 h)
`V1–V28`	numeric	Principal components of confidential original features (anonymised by the bank)
`Amount`	numeric	Transaction amount in euros
`Class`	binary	1 = fraud, 0 = legitimate (used only for evaluation and to remove frauds from training)

3.2 Data quality

The data has no missing values and all columns are numeric, so no imputation is needed. We found 1,081 exact duplicate rows, 19 of them frauds. We drop them *before* splitting, because an identical row in both the training and the test block leaks information and inflates test scores. After deduplication 283,726 rows remain, including 473 frauds.

3.3 Class imbalance

Fraud makes up only 0.167% of the rows, about one transaction in 600 (Fig. 1). A model that never flags anything would be 99.83% accurate and would catch no fraud. Accuracy is therefore meaningless here. We use the triage metrics defined in Section 5.8.

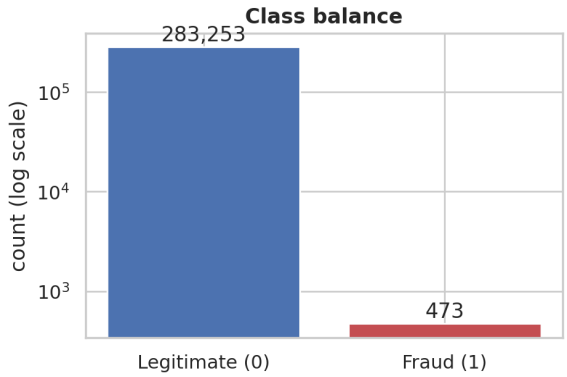


Figure 1: Class balance on a logarithmic scale: legitimate transactions outnumber frauds by about 600 to 1.

3.4 Time of day and transaction amount

Transaction volume collapses in the early hours of the day, while the *fraud rate* peaks there, at up to about nine times the average (Fig. 2). The raw `Time` value only encodes a row's position within the two-day file, so we keep the hour-of-day signal instead (Section 4). `Amount` is extremely right-skewed (skewness > 15). Frauds have a lower median amount than legitimate transactions, because many are small "test" charges, but they also have a long tail (Fig. 3). Amount alone cannot separate the classes.

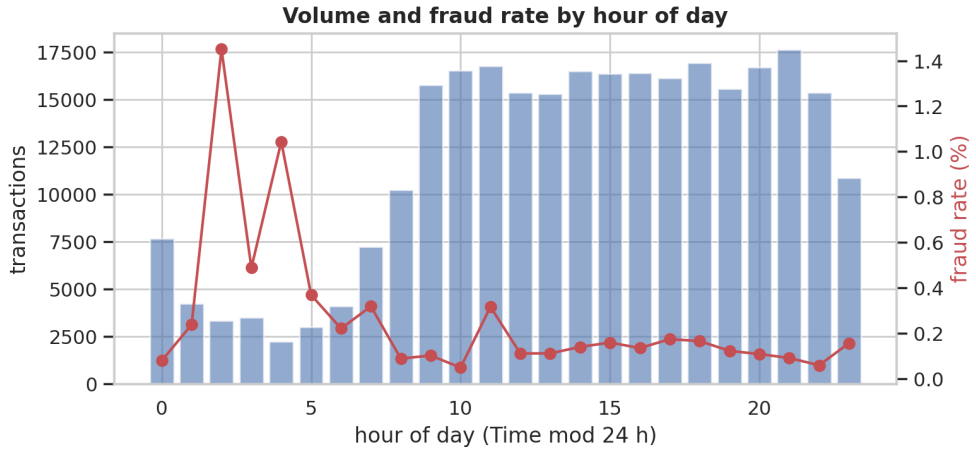


Figure 2: Transaction volume (bars) and fraud rate (line) by hour of day. Fraud is rare in absolute terms but proportionally much more frequent at night.

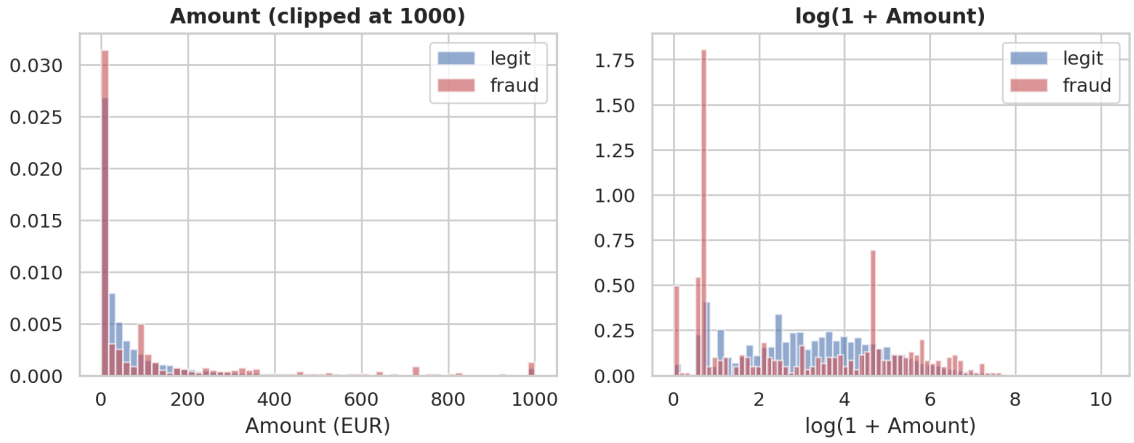


Figure 3: Distribution of the transaction amount (left, clipped at 1,000) and of $\log(1 + \text{Amount})$ (right) for legitimate and fraudulent transactions.

3.5 Which components separate the classes?

Labels are used here *only for analysis*; the models never see them during training. We rank features by the two-sample Kolmogorov–Smirnov (KS) statistic between the legitimate and the fraud distributions (Fig. 4). The strongest separators are V14 (0.84), V10 (0.80), V12 (0.78), V4 (0.76), V11 (0.75), V17 (0.74). Fig. 5 shows that frauds sit in the *tails* of these components,

far from where legitimate transactions concentrate. This is exactly the structure a density-based detector exploits: a transaction in a low-density region of "normal" space is suspicious.

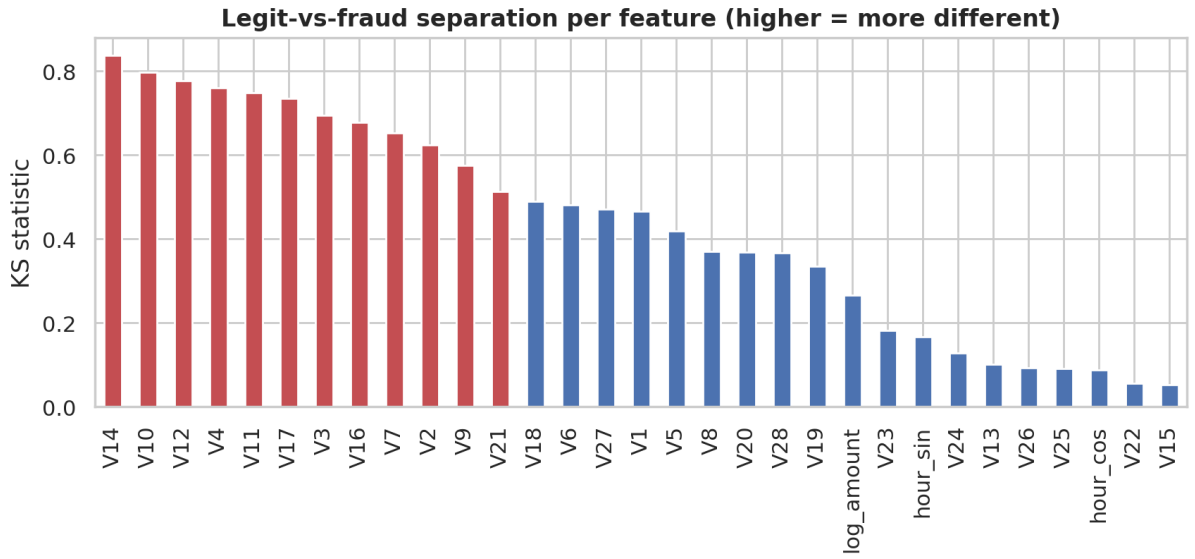


Figure 4: Kolmogorov–Smirnov separation between legitimate and fraud distributions, per feature (higher = more different).

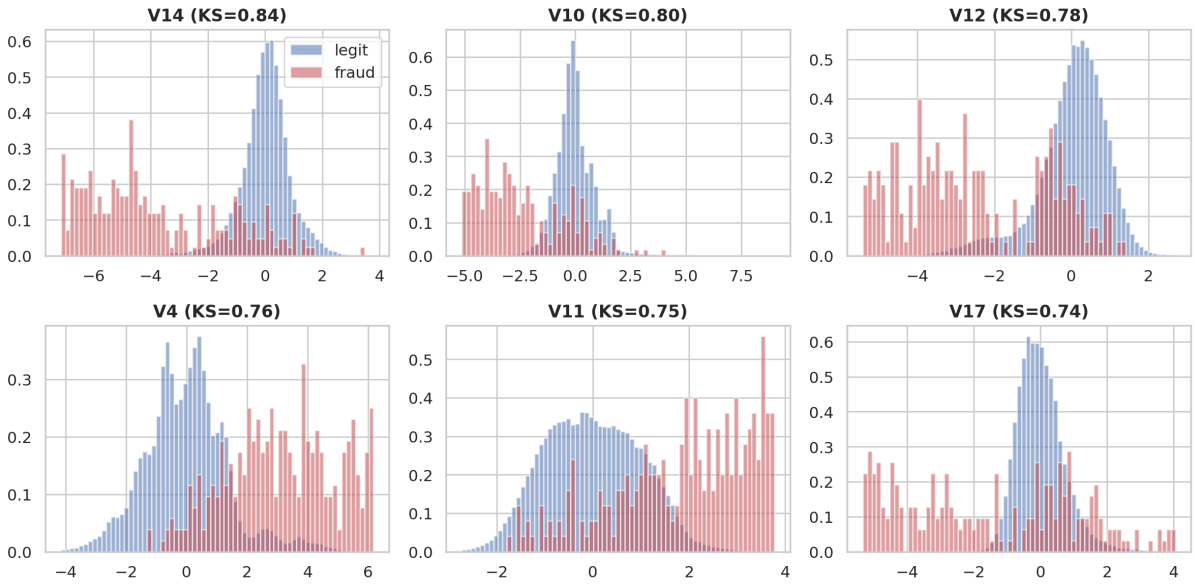


Figure 5: Densities of the six most separating components. Frauds (red) occupy the tails of the legitimate distribution (blue).

3.6 Is legitimate behaviour Gaussian?

A Gaussian density assumes symmetric, light-tailed marginals (skewness 0, excess kurtosis 0). On legitimate rows, 13 of the 31 model features have |skewness| > 1 and 18 have excess kurtosis above 3. The extreme case is V28, with excess kurtosis of about 966. The Q–Q plots in Fig. 6 bend strongly away from the Gaussian line. A single Gaussian is therefore a poor fit, and even

mixture components are strained. This quantifies the gap named in the brief: the Gaussian-mixture assumption may not hold for the legitimate transaction manifold.

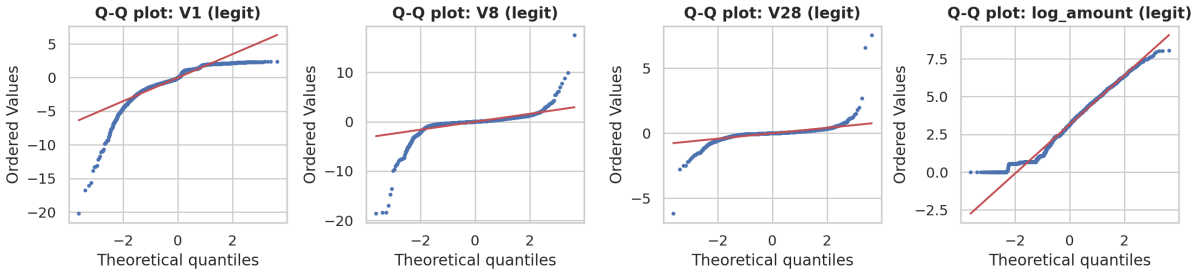


Figure 6: Q–Q plots of legitimate transactions against a normal distribution. Points far from the line indicate heavy tails and skew.

3.7 Correlation structure

Because V1–V28 are principal components, they are nearly uncorrelated over the whole dataset. *Within the fraud class*, however, they are strongly correlated (Fig. 7): fraud moves several components together. A full-covariance density model can recognise such an unusual *combination* even when no single feature is extreme, which a per-feature threshold rule could not.

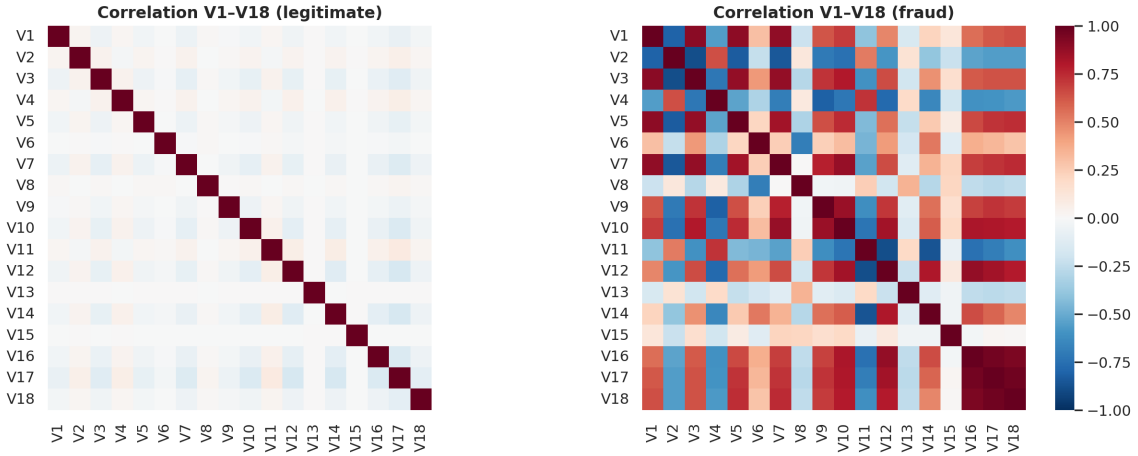


Figure 7: Correlation of V1–V18 among legitimate transactions (left) and among frauds (right).

3.8 Chronological split and base rates

Rows are ordered by `Time` and cut into three consecutive blocks (Table 3, Fig. 8). Models are trained on the past and evaluated on the future, as they would be in production. The scaler and the GMM are fitted only on the 169,894 legitimate training rows. The fraud label is used once, to remove frauds from that block. The validation block (mixed) is used to choose hyper-parameters and thresholds. The test block (mixed) is used once, at the end. Base rates drift between blocks, so every precision value must be read against its block's base rate.

Table 3: Chronological split and base rates.

Block	Rows	Frauds	Fraud rate	Used for
Train (first 60%)	170,236	342	0.201%	fitting scaler and GMM (legitimate rows only)
Validation (next 20%)	56,745	57	0.100%	choosing hyper-parameters and thresholds
Test (last 20%)	56,745	74	0.130%	final evaluation, used once

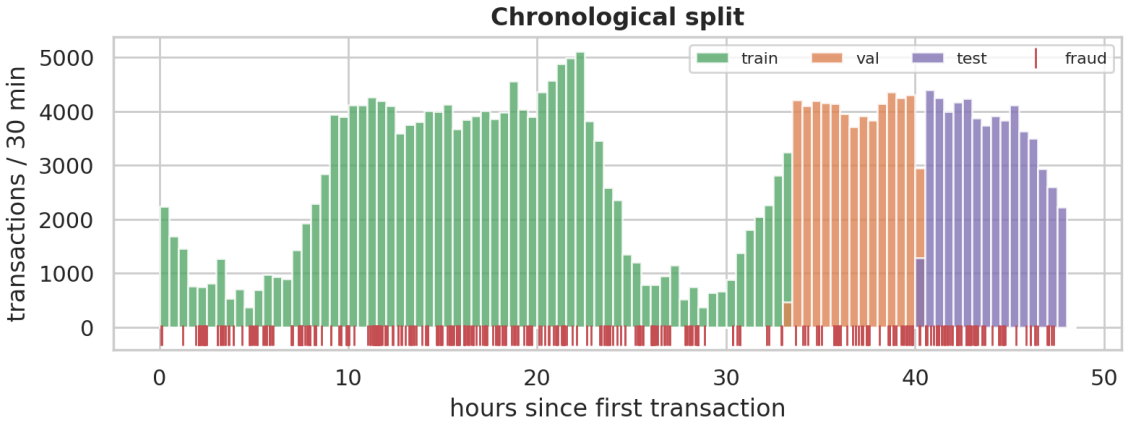


Figure 8: Transactions per 30 minutes, coloured by block; red ticks mark frauds.

Limitation. The validation and test blocks (hours 33–48) cover the daytime and evening of day two. The night-time hours, where the fraud *rate* peaks, occur only in the training block, so the time-of-day feature is evaluated on daytime transactions only.

4 Feature Engineering

4.1 Engineered features

Table 4 lists every transformation and its rationale. The hour of day is placed on the unit circle, so that 23:00 and 00:00 are neighbours:

$$h = \frac{\texttt{Time} \bmod 86400}{3600}, \qquad \text{hour_sin} = \sin\left(\frac{2\pi h}{24}\right), \qquad \text{hour_cos} = \cos\left(\frac{2\pi h}{24}\right). \tag{1}$$

Table 4: Feature-engineering decisions.

Item	Choice	Reason
Duplicates	dropped before splitting	avoid train/test leakage
`Time`	→ hour_sin, hour_cos; raw value dropped	keep the night-time fraud signal, remove the position-in-file artefact
`Amount`	→ log(1 + Amount)	remove extreme skew so one feature cannot dominate
V1–V28	kept	already decorrelated principal components
Scaling	fitted on legitimate training rows only	no information from validation, test or fraud
Labels	used only to drop training frauds, set thresholds, score test	"keep the fraud label hidden during unsupervised training"

We compare three feature sets: *pca* (V1–V28, d=28), *pca+amount* (d=29) and *full* (d=31, adding the hour features).

4.2 Scaling options

Four scalers are compared, each fitted on legitimate training rows only:

- **Standard:** $z = (x - \mu)/\sigma$.
- **Robust:** $z = (x - \text{median})/\text{IQR}$, which is less sensitive to outliers.
- **Power (Yeo–Johnson):** a parametric transform that reduces skew, followed by standardisation.
- **Quantile:** $z = \Phi^{-1}(\hat{F}(x))$, where $\hat{F}$ is the empirical CDF of the feature on legitimate training rows and Φ is the standard normal CDF. Every *marginal* becomes exactly standard normal on the training data. Values beyond the training range are clipped to the extreme quantiles.

5 Methodology

5.1 Overview

Fig. 9 summarises the pipeline. Everything to the left of the scoring step uses legitimate training rows only.

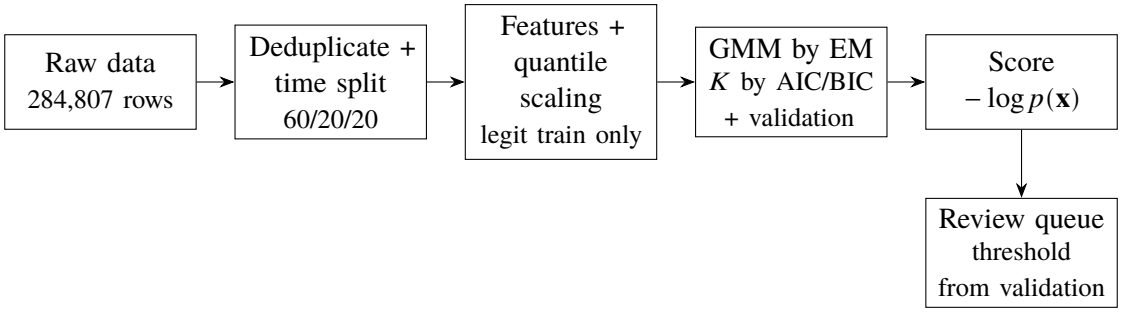


Figure 9: Phase-1 pipeline. The model sees only legitimate training transactions; labels are used only for choosing thresholds (validation) and for the final evaluation (test).

5.2 Gaussian Mixture Model

A GMM with K components models the density of legitimate transactions as

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1, \tag{2}$$

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-d/2} \, |\boldsymbol{\Sigma}|^{-1/2} \exp\Big(-\tfrac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \Big). \tag{3}$$

Each component can be read as one "kind" of normal transaction. The mixture can represent multi-modal and skewed shapes, but every component is an ellipsoid.

5.3 Expectation–Maximisation

Given N legitimate training rows, the log-likelihood $\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ has no closed-form maximiser, because the logarithm of a sum couples the components. EM introduces the unobserved component membership of each row and alternates two steps (Algorithm 1).

E-step. Compute the *responsibility* of component k for row i:

$$\gamma_{ik} = \frac{\pi_k \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}. \tag{4}$$

M-step. With $N_k = \sum_i \gamma_{ik}$, re-estimate the parameters as responsibility-weighted averages:

$$\pi_k = \frac{N_k}{N}, \qquad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_i \gamma_{ik} \mathbf{x}_i, \qquad \boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_i \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^{\top} + \lambda \mathbf{I}. \tag{5}$$

Each EM iteration never decreases $\ell(\theta)$ [8], but EM converges only to a *local* optimum that depends on the initialisation. We initialise with k-means++ and study the effect of the random seed (Section 6.7). The ridge term $\lambda \mathbf{I}$ (`reg_covar`, default 10^{-4}) keeps every covariance invertible when a component collapses onto a few nearly identical points; without it the likelihood can diverge to infinity.

Algorithm 1 EM for a Gaussian mixture fitted on legitimate transactions

Require: legitimate training rows $\mathbf{x}_1, \dots, \mathbf{x}_N$; number of components K; regulariser λ

Initialise $\boldsymbol{\mu}_k$ with k-means++; set $\pi_k = 1/K$ and $\boldsymbol{\Sigma}_k$ from the k-means clusters
repeat
  E-step: $\gamma_{ik} \leftarrow \pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \,/\, \sum_j \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$
  M-step: $N_k \leftarrow \sum_i \gamma_{ik}$; $\pi_k \leftarrow N_k/N$; $\boldsymbol{\mu}_k \leftarrow \frac{1}{N_k} \sum_i \gamma_{ik} \mathbf{x}_i$
    $\boldsymbol{\Sigma}_k \leftarrow \frac{1}{N_k} \sum_i \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top + \lambda \mathbf{I}$
until the increase of $\ell(\theta)$ is below a tolerance (or 500 iterations)
return $\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$; anomaly score $s(\mathbf{x}) = -\log p(\mathbf{x})$

5.4 Anomaly score and decision rule

The anomaly score is the negative log-likelihood under the legitimate density,

$$s(\mathbf{x}) = -\log p(\mathbf{x}), \tag{6}$$

so a transaction in a low-density region of legitimate behaviour receives a high score. A deployed system raises an alert when $s(\mathbf{x}) > t$. The threshold t is the $(1 - f)$-quantile of the scores of legitimate *validation* transactions, where f is the false-positive budget. It is then frozen and applied to the test period. For $K = 1$ the score reduces (up to constants) to the squared Mahalanobis distance $(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$, which we use as a baseline.

5.5 Model selection with AIC and BIC

With p free parameters, maximised likelihood $\hat{L}$ and n rows,

$$\mathrm{AIC} = 2p - 2\log \hat{L}, \qquad \mathrm{BIC} = p \log n - 2 \log \hat{L}. \tag{7}$$

Both reward fit and penalise complexity. BIC penalises more strongly ($\log n > 2$ for $n > 7$), so it prefers smaller models. Table 5 gives the parameter counts.

Table 5: Free parameters of a K-component GMM in d dimensions.

Covariance	**Free parameters** p	**Per component,** $d = 28$
Full	$(K - 1) + Kd + K\, d(d + 1)/2$	434
Diagonal	$(K - 1) + 2Kd$	$2d = 56$

5.6 Computational complexity

One EM iteration with full covariances costs $\mathcal{O}(NKd^2)$ for the E-step, which evaluates every Mahalanobis form, plus $\mathcal{O}(Kd^3)$ for the Cholesky factorisations. For $N \approx 1.7 \times 10^5$, $d = 28$ and $K \le 24$ a fit takes seconds on a laptop CPU, but more than a minute on Colab's free CPU. The selection studies (about 100 fits) therefore use a random 40,000-row *development subsample* of the legitimate training rows. The final model and all baselines use all rows.

5.7 Reference baselines

To put the GMM's numbers in context we evaluate:

- a **single Gaussian** ($K = 1$, Mahalanobis distance) on the same inputs;
- the same single Gaussian on **standard-scaled** inputs, which isolates the effect of feature engineering;
- the **BIC-selected** mixture;
- **Isolation Forest** [7] (300 trees), fitted on legitimate rows only;
- a **supervised logistic regression** (class-balanced, $C = 0.1$) that *uses fraud labels*. This is an upper reference showing what labels are worth, not a fair competitor.

5.8 Evaluation protocol and metrics

All scores follow the convention "higher = more suspicious". Let $y_i \in \{0, 1\}$ be the label and s_i the score.

- **PR-AUC (average precision).** $\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n$ over all thresholds, where P_n and R_n are precision and recall. A random ranking scores the base rate (0.130% on test). We add a 95% stratified bootstrap confidence interval (1,000 resamples).
- **Recall at a false-positive budget** $f \in \{0.1\%, 0.5\%, 1\%\}$**.** The threshold is set on validation so that a fraction f of legitimate validation transactions would be flagged. We report test recall and the *realised* test false-positive rate.
- **Precision at a review capacity** $k \in \{50, 100, 200, 500\}$ **(P@k).** The share of fraud among the k highest-scored test transactions.
- **ROC-AUC** as a secondary metric only.

Because EM depends on its initialisation, every comparison during development is the mean over three seeds.

5.9 Experimental setup

Table 6: Experimental setup and hyper-parameters.

Item	Setting
Feature-set / scaler grid	{pca, pca+amount, full} × {standard, robust, power, quantile}; GMM full, $K = 4$, 3 seeds
K grid	{1, 2, 3, 4, 6, 8, 12, 16, 24}; full (3 seeds) and diagonal (1 seed) covariances
EM	k-means++ initialisation, $\lambda = 10^{-4}$, max. 500 iterations; final model with 5 restarts (`n_init=5`)
Selection rule	AIC/BIC reported; K chosen by mean validation PR-AUC
Stability study	5 EM seeds for $K \in \{4, 8, 16\}$, quantile vs. standard inputs
Bootstrap	1,000 stratified resamples for confidence intervals
Software	Python 3, scikit-learn, NumPy, pandas, SciPy, matplotlib [15]
Hardware	Google Colab (standard CPU runtime); global seed 42

6 Experiments and Results

6.1 Feature-engineering ablation

Table 7 and Fig. 10 compare feature sets and scalers with a fixed GMM (full covariance, $K = 4$). **Quantile scaling is the best scaler for every feature set, and it has by far the smallest seed-to-seed spread.** With standard, robust or power scaling, the heavy tails of legitimate features (Section 3) let a few extreme legitimate values dominate the covariance estimates. Different EM runs then lock onto different extreme points, and validation PR-AUC swings by $\pm$0.1–0.3. The selected inputs are *pca + quantile*. With only 57 validation frauds, the differences between feature sets under quantile scaling are within the noise level; the decision that matters is the scaler.

Table 7: Feature ablation: validation PR-AUC (mean $\pm$ std over 3 EM seeds; GMM, full covariance, $K = 4$). Best in bold.

Feature set	standard	robust	power	quantile
pca	0.296$\pm$0.290	0.578$\pm$0.168	0.085$\pm$0.010	**0.698$\pm$0.009**
pca+amount	0.246$\pm$0.133	0.165$\pm$0.121	0.453$\pm$0.114	0.659$\pm$0.021
full	0.507$\pm$0.291	0.468$\pm$0.212	0.096$\pm$0.024	0.631$\pm$0.086

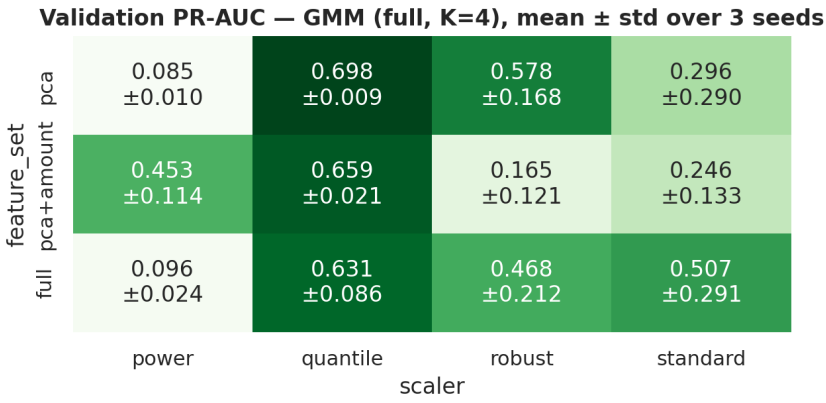


Figure 10: Feature ablation heat map (validation PR-AUC, mean ± std over 3 seeds).

6.2 Choosing K: information criteria versus detection quality

Fig. 11 and Table 8 show the sweep on the selected inputs.

AIC, BIC and the label-free held-out log-likelihood *all keep improving* up to the largest model tried ($K = 24$). With tens of thousands of rows, the penalty $p \log n$ is small next to the likelihood gained by describing the heavy tails ever more finely. Detection quality does not follow. Validation PR-AUC is flat or lower for large K, and its seed-to-seed spread grows, because more components mean more local optima and extra components begin to model rare legitimate behaviour in the sparse regions where fraud also lives. Diagonal covariances are far worse at every K (best 0.108 at $K = 24$): fraud is visible in *combinations* of components, which a diagonal model cannot represent.

We therefore report the BIC choice as a baseline, but select K by mean validation PR-AUC, which is the purpose of the validation block. The final model (full covariance, $K = 4$) is refit on *all* legitimate training rows with five EM restarts, keeping the restart with the highest likelihood (a label-free choice).

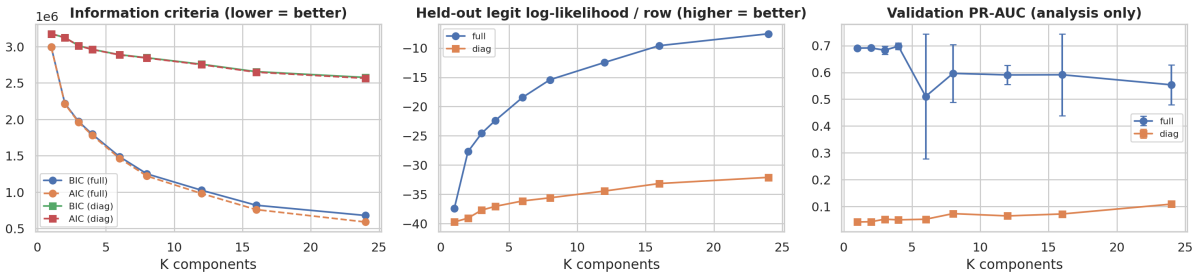


Figure 11: Model selection on quantile-scaled inputs. Left: AIC and BIC keep decreasing with K. Centre: held-out legitimate log-likelihood keeps increasing. Right: validation PR-AUC (mean ± std over 3 seeds) does not improve with K.

Table 8: Full-covariance sweep on the development subsample (BIC in millions; PR-AUC mean ± std over 3 seeds).

K	Parameters	BIC	Held-out ℓ/row	Validation PR-AUC
1	434	2.995	-37.42	0.691±0.000
2	869	2.218	-27.71	0.691±0.001
3	1,304	1.971	-24.58	0.683±0.015
4	1,739	1.795	-22.38	0.698±0.012
6	2,609	1.486	-18.43	0.510±0.233
8	3,479	1.251	-15.42	0.596±0.107
12	5,219	1.025	-12.46	0.591±0.036
16	6,959	0.819	-9.60	0.591±0.153
24	10,439	0.678	-7.56	0.554±0.074

6.3 Main results

Table 9 reports test performance with thresholds fixed on validation, and Fig. 12 shows the curves.

Table 9: Test results (thresholds for the recall columns chosen on validation and frozen). P@k = precision of the top-k alerts. Base rate on test = 0.130%.

Model	PR-AUC	95% CI	ROC-AUC	Rec@0.1%	Rec@0.5%	Rec@1%	P@50	P@100
GMM (K=4, full)	0.686	[0.57, 0.81]	0.934	0.757	0.784	0.784	0.90	0.56
GMM, BIC choice (K=24, full)	0.626	[0.51, 0.76]	0.955	0.662	0.757	0.770	0.86	0.53
Single Gaussian (K=1)	0.691	[0.58, 0.80]	0.963	0.757	0.757	0.770	0.92	0.56
Single Gaussian, standard scaling	0.163	[0.12, 0.22]	0.959	0.081	0.514	0.716	0.18	0.28
Isolation Forest	0.050	[0.04, 0.07]	0.939	0.014	0.203	0.432	0.02	0.06
Supervised LogReg (uses labels)	0.739	[0.65, 0.83]	0.981	0.770	0.851	0.851	0.88	0.56

The main observations are:

- The selected GMM reaches **PR-AUC 0.686**. At the strictest budget (0.1% of legitimate customers flagged on validation) it catches **75.7%** of test frauds with 94 alerts, of which 59.6% are fraud. Its top-50 queue is **90% fraud**.
- **Mixture size barely matters:** a single Gaussian on the same inputs scores 0.691; the BIC-selected mixture ($K = 24$) scores 0.626.
- **The scaler matters enormously:** the same single Gaussian on standard-scaled inputs collapses to 0.163.
- Isolation Forest reaches only 0.050. The label-using logistic regression (upper reference) reaches 0.739, so the label-free GMM recovers most of what labels are worth.

ROC-AUC is high (0.93–0.98) for almost every model, including those with poor precision. This confirms that ROC is uninformative under this imbalance [14].

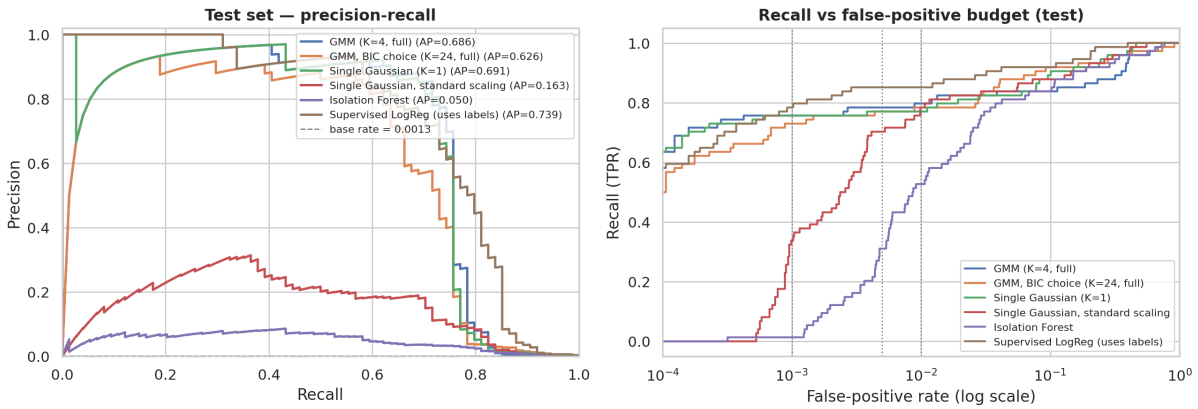


Figure 12: Test precision–recall curves (left) and recall versus false-positive rate on a log scale (right). Dotted lines mark the 0.1%, 0.5% and 1% budgets.

6.4 Score distribution

Fig. 13 shows the distribution of the GMM score on the test set. Most frauds receive scores far beyond the bulk of legitimate transactions, while a minority overlap with legitimate behaviour. These overlapping frauds cannot be caught without flagging many legitimate customers.

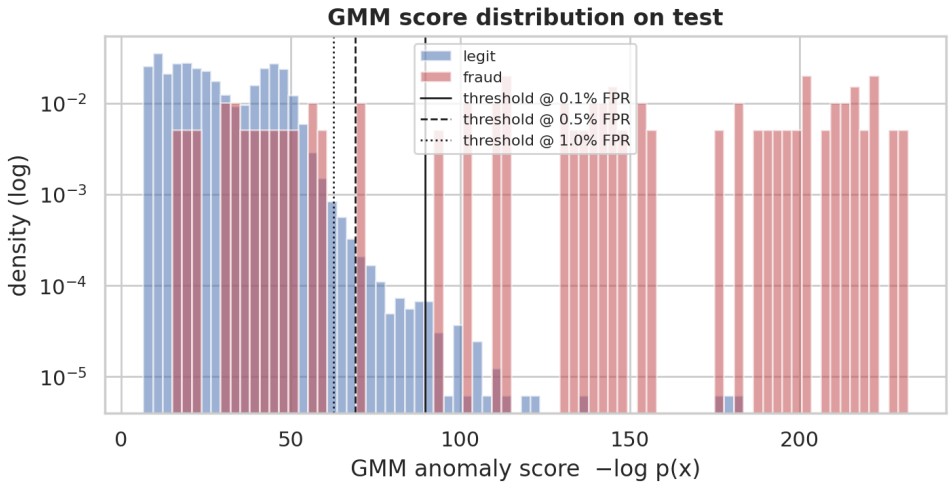


Figure 13: GMM anomaly score on test (log density axis) with the three validation-chosen thresholds.

6.5 Threshold analysis and false-positive trade-offs

Table 10 turns the scores into operating policies for the 7.7-hour test window. Tightening the budget from 1% to 0.1% shrinks the queue from 399 to 94 alerts while losing only a couple of frauds. Most frauds are either very anomalous (caught at any budget) or look entirely normal (missed at every budget), so the recall curve is flat between budgets (Fig. 14). The realised test false-positive rate stays below its validation target because the test period is slightly "cleaner" than the validation period.

Table 10: GMM operating points on the test period (thresholds from validation).

FP budget	Alerts	Alerts / hour	Frauds caught	Precision	Realised FPR
0.1%	94	12.3	56/74	0.596	0.067%
0.5%	221	28.9	58/74	0.262	0.288%
1.0%	399	52.1	58/74	0.145	0.602%

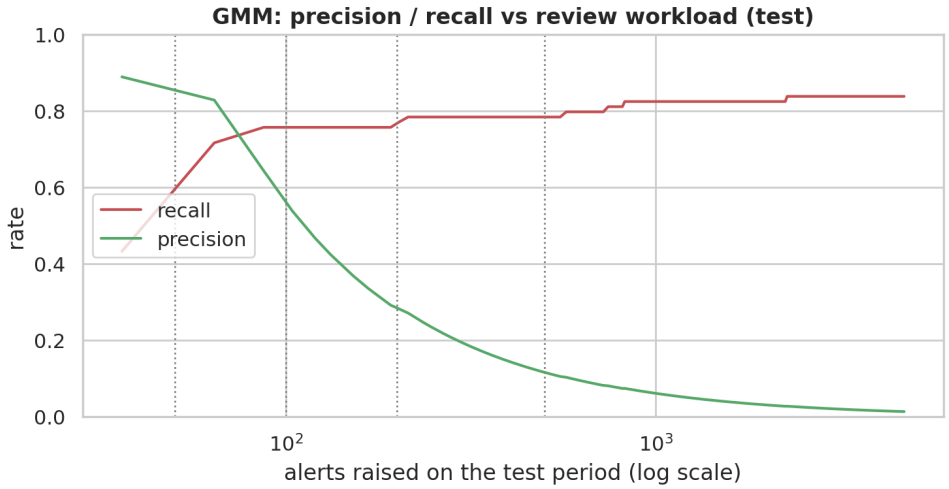


Figure 14: Precision and recall versus review workload (number of alerts, log scale) on the test period.

6.6 Cost-based operating point

A simple cost model makes the trade-off concrete. Assume each alert costs a fixed review cost c, and each missed fraud loses its transaction amount. Reviewing the top k alerts then costs

$$\text{Cost}(k) = c \cdot k + \sum_{i \in \text{frauds not in top-}k} \text{Amount}_i. \tag{8}$$

With $c = 5$, the cost is minimised by reviewing the top 61 alerts, for a total of 2,763, compared with 7,728 if no model is used (Fig. 15). The cost values are illustrative, but the method lets a bank choose its operating point from its own review and loss costs.

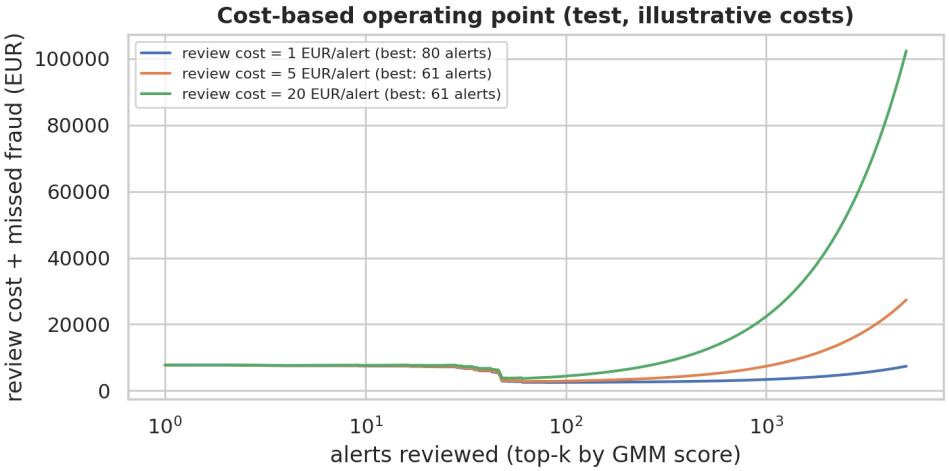


Figure 15: Review cost plus missed-fraud amount versus the number of alerts reviewed, for three review costs per alert.

6.7 Stability: the high-dimensional gap, measured

Table 11 and Fig. 16 refit the GMM with five EM seeds. With standard scaling at $K = 4$, test PR-AUC ranges from 0.148 to 0.743 (std 0.224): the same model, started differently, is either useful or useless. With quantile inputs the range narrows to 0.652–0.737 (std 0.032), and the rankings and top-100 queues agree far more. Instability grows with K for both input types. Even for the final model, the fitted covariance matrices have condition numbers up to about 2,467, so several directions are almost flat. This shows how close a 28-dimensional full-covariance mixture is to degeneracy. Moderate regularisation (λ between 0 and 10^{-2}) barely changes validation PR-AUC; strong regularisation ($\lambda = 0.1$) blurs the density and hurts.

Table 11: Seed stability on test (5 EM seeds): PR-AUC mean $\pm$ std and range, mean Spearman ρ between seeds, and mean overlap of the top-100 alerts.

K	Inputs	PR-AUC	Range	Spearman ρ	Top-100 overlap
4	quantile (selected)	0.705±0.032	[0.65, 0.74]	0.897	0.87
4	standard	0.505±0.224	[0.15, 0.74]	0.851	0.77
8	quantile (selected)	0.650±0.033	[0.59, 0.69]	0.866	0.79
8	standard	0.418±0.226	[0.11, 0.68]	0.798	0.63
16	quantile (selected)	0.604±0.089	[0.49, 0.74]	0.906	0.74
16	standard	0.511±0.229	[0.10, 0.72]	0.813	0.63

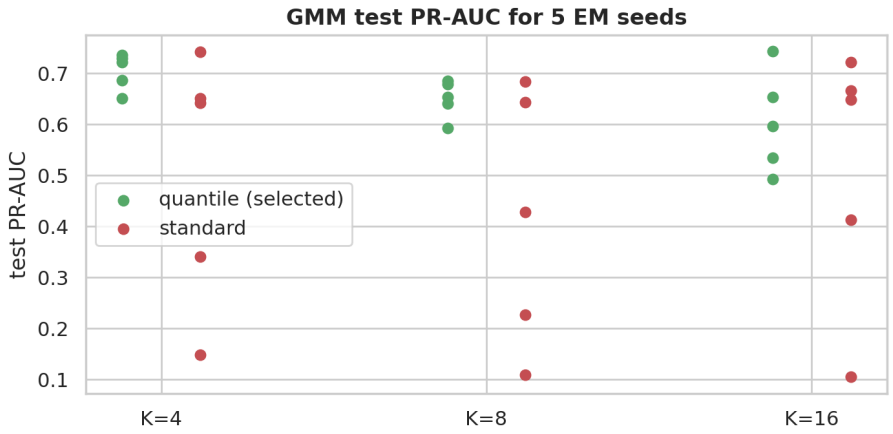


Figure 16: Test PR-AUC of five EM seeds per configuration: quantile inputs (green) versus standard scaling (red).

6.8 Error analysis

At the 0.5% budget the caught frauds have extreme values in V14 and V17 (mean $|V14| \approx 8$), while the missed frauds look ordinary in these components (about a quarter of that) and sit in dense legitimate regions. False alarms are legitimate transactions in sparse regions: unusual component values and larger amounts. They are "rare but honest" behaviour, which any purely density-based detector will flag.

7 Discussion

7.1 Interpretation

The results show that, for density-based fraud detection, the *representation* matters more than the model's flexibility. The quantile transform moves every feature onto a common Gaussian scale. It removes the heavy tails that otherwise dominate the covariance estimates, and it makes the joint structure of legitimate behaviour close enough to Gaussian that even one component works well. Information criteria answer a different question ("which model describes legitimate data best?") from the one we care about ("which model ranks fraud highest?"). At this sample size they keep favouring larger mixtures that do not detect fraud better.

7.2 The gap that motivates Phase 2

1. **Shape assumption.** The GMM works only after every marginal is *forced* to look Gaussian, and the clipping in that transform discards information about how extreme a value is. The joint structure remains non-Gaussian.
2. **Instability in high dimensions.** Each component costs 434 parameters in 28 dimensions. EM's solution depends on its seed, and the covariances are near-singular.
3. **Likelihood $\neq$ detection.** Unsupervised model selection (AIC/BIC) keeps asking for

larger models that detect fraud no better and are less stable.

4. **Mixture adds little.** $K = 1$, $K = 4$ and $K = 24$ perform alike. Adding ellipsoids does not capture the curved, low-dimensional structure of legitimate behaviour.

Phase 2 replaces the fixed Gaussian shape with a *learned non-linear* model of legitimate behaviour (undercomplete and denoising autoencoders [16,17]). It then tests hybrids that combine the Gaussian density with the autoencoder, in the spirit of DAGMM [18].

7.3 Limitations and threats to validity

- **Short horizon.** The data covers two days in 2013; long-term concept drift is untested.
- **Anonymised features.** PCA anonymisation prevents domain features (merchant, customer history, device).
- **Daytime test period.** Validation and test fall in daytime hours (Section 3).
- **Few test frauds.** With 74 test frauds, confidence intervals are about ± 0.1 PR-AUC wide; differences smaller than this should not be over-interpreted.
- **Development subsample.** Selection studies use a 40,000-row subsample; the final model uses all rows.

8 Conclusion and Future Work

A Gaussian-mixture density of legitimate transactions, fitted without any fraud label, ranks fraud well enough for practical triage. It reaches a test PR-AUC of 0.686, catches 75.7% of frauds at a 0.1% false-positive budget, and fills a top-50 review queue that is 90% fraud. The decisive ingredient was feature engineering (marginal Gaussianisation), not the number of components. The information criteria the textbook recommends keep asking for larger models that bring no detection gain at this sample size. Together with the measured EM instability, these findings define the gap addressed in Phase 2: learning a non-linear model of legitimate behaviour with autoencoders, and testing whether a hybrid of the two paradigms can do better than either alone.

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A Reproducibility

- **Code.** Four self-contained notebooks (`notebooks/01–04`) run top-to-bottom on Google Colab. Phase 1 uses notebooks 01 (data and features) and 02 (GMM). Each notebook downloads the dataset from a public mirror and verifies its SHA-256 checksum.
- **Seeds.** Global seed 42, plus explicit seed lists for multi-seed studies.
- **Traceability.** Every number in this report is generated automatically from the notebooks' saved results (`results/*.json`) into `numbers.tex`. Every figure is saved by the notebooks into `figures/`.
- **Run time on Colab.** Notebook 01 takes about 2 minutes and notebook 02 about 35 minutes (CPU runtime).

B Team Contributions

Table 12: Team T068 (Section B): members and primary responsibilities. Every member contributed to the analysis and can explain the full pipeline.

Name	Enrollment No.	Primary responsibility (Phase 1)
Sharma Piyush	2401010437	Data quality, exploratory analysis, chronological split and leakage control
Saransh Singh Tomar	2401010418	Feature engineering and scaling ablation (quantile transform, cyclic time, log amount)
Satwik Mani Tripathi	2401020056	GMM theory, EM algorithm, AIC/BIC model selection and stability study
Gourav Tanwar	2401010173	Reference baselines (single Gaussian, Isolation Forest, supervised reference) and error analysis
Alok Kumar	2401010058	Evaluation protocol, threshold and cost analysis, report and presentation